

Astro HW 11

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11.1

a) Spiral galaxies - Tracing the Light

$$L(r) = \int_0^r 2\pi s \Sigma(s) ds = 2\pi h_r^2 \left[1 - \left(\frac{r}{1 + h_r} \right) e^{-r/h_r} \right]$$

$$L(r) = \int_0^R 2\pi r [\Sigma_0 \exp(-r/h_r)] dr$$

$$L(r) = 2\pi \Sigma_0 \int_0^R r [\exp(-r/h_r)] dr$$

$$x = r/h_r, \quad x h_r = r, \quad dx h_r = dr$$

$$= 2\pi \Sigma_0 h_r^2 \int_0^R x e^{-x} dx$$

$$= 2\pi \Sigma_0 h_r^2 \left[\frac{R}{h_r} - (R/h_r + 1)e^{-R/h_r} + C \right]$$

$$L(r) = 2\pi \Sigma_0 h_r^2 [1 - (R/h_r + 1)e^{-R/h_r}]$$

b)

Given ...

$$\Sigma = 36 L_{\odot} \text{pc}^{-2}, \quad h_r = 4 \text{kpc}, \quad R_0 = 8.5 \text{kpc}$$

$$\Sigma = \Sigma_0 \exp(-r/h_r)$$

We find ...

$$\Sigma_0 = 301.42 L_{\odot} \text{pc}^{-2}$$

c)

$$L(r) = 2\pi\Sigma_0 h_r^2 \left[1 - \frac{(R/h_r + 1)}{e^{R/h_r}} \right]$$

$$L(R) = 2\pi\Sigma_0 h_r^2 \lim_{R \rightarrow \infty} \left[\frac{(R/h_r)}{e^{R/h_r}} \right]$$

Apply L'hoptials ...

$$L(\infty) = 2\pi\Sigma_0 h_r^2 \lim_{R \rightarrow \infty} \left[\frac{(1/h_r)}{R e^{R/h_r}} \right]$$

$$L(\infty) = 2\pi\Sigma_0 h_r^2$$

$$\boxed{L(\infty) \approx 3.030 \times 10^{10} \text{ L}_\odot}$$

d)

Given...

$$L/3 = M_{mw}, \quad L = 3.030 \times 10^{10} \text{ L}_\odot$$

We find ...

$$\boxed{M_{mw} \approx 1.010 \times 10^{10} \text{ M}_\odot}$$

e)

Given ...

$$L(R_0)/L(\infty), \quad L(\infty) = 3.030 \times 10^{10} \text{ L}_\odot$$

$$L(r) = 2\pi\Sigma_0 h_r^2 [1 - (R/h_r + 1)] e^{-R/h_r}$$

We find ...

$$L(R_0) = 1.899 \times 10^{10} \text{ L}_\odot \text{ pc}^{-2}$$

$$L(R_0)/L(\infty) = 62\%$$

11.2 Spiral galaxies - Tracing the Matter

a)

Given...

$$v_{esc} = \sqrt{\frac{2GM}{R}}$$

$$v_{esc} = 500 \text{ km s}^{-1} \quad 8.5 \text{ kpc}$$

We find that...

$$v_{esc} = \frac{\sqrt{2GM}}{\sqrt{R}}$$

$$\frac{v_{esc} \sqrt{R}}{\sqrt{2G}} = \sqrt{M}$$

$$M = 2.470 \times 10^{11} \text{ M}_\odot$$

b)

i)

Given ...

$$M_V = -2.76 \log_{10} \left(\frac{P}{10\text{d}} \right) - 4.16$$

$$m_V = 18.9, \quad P = 31.47\text{d}$$

$$m - M = 5 \log_{10} \left(\frac{d}{10\text{pc}} \right) \rightarrow d = 10 \exp \left(\frac{m - M + 5}{5} \right) \text{pc}$$

We find ...

$$\boxed{M_v = -5.534, \quad d = 7.706 \times 10^{-1} \text{Mpc}}$$

ii)

Given ...

$$v = H_0 d$$

$$d = d = 7.706 \times 10^{-1} \text{Mpc}$$

We find ...

$$\boxed{53.9\text{km/s}}$$

iii)

Given ...

$$\frac{\Delta v}{\Delta T} = a$$

$$v_{init} = 53.9\text{km/s}, \quad v_{final} = -119\text{km/s}, \quad \Delta v = v_{final} - v_{init} \quad \Delta T = 14\text{Gyr}$$

$$\boxed{g = -3.914 \times 10^{-13} \text{m/s}}$$

iv)

Given ...

$$a = GM/R^2$$

$$g_{M31} = GM_{m31}/R^2, \quad g_{MW} = GM_{MW}/R^2$$

$$R = 7.706 \times 10^{-1} \text{Mpc}$$

$$a = a_m + a_{m31}$$

We can find ...

$$\cancel{3.914 \times 10^{-13}} = \cancel{3} GM_{MW}/R^2$$

$$\frac{(3.914 \times 10^{-13})R^2}{3G} = M_{MW}$$

$$M_{MW} = 5.559 \times 10^{11} \text{M}_{\odot}$$

11.3

a)

Given ...

$$R = D\theta$$

$$D = 16.8 \text{ Mpc}, \quad \theta = .1''$$

We find ...

$$R = 8.144 \text{ pc}$$

b)

From figure ...

$$-.1'' = 1780 \text{ km/s}$$

$$+.1'' = 650 \text{ km/s}$$

We find ...

$$c_{vel} = 1215 \text{ km/s}$$

c)

Given ...

Assuming circular orbit ...

$$v^2 = \frac{GM}{R}$$

$$M = \frac{v^2 R}{G}$$

$$R = 8.144 \text{ pc}$$

We find ...

$$M_{bh} = 4.613 \times 10^{38} \text{ kg} = 2.320 \times 10^8 M_{\odot}$$

11.4 Redshifts and the Hubble Law

a)

Given ...

$$Z = \frac{\lambda_{obsv} - \lambda_{rest}}{\lambda_{rest}}$$

$$\lambda_{rest} = 486.135 \text{ nm}, \quad \lambda_{obsv} = 497.97 \text{ nm}$$

We find ...

$$Z = .0023$$

b)

Given ...

$$\begin{aligned}m - M &= 5 \log(d/10\text{pc}) \\ v = H_0 d, \quad z &\simeq \frac{v}{c}, \quad \rightarrow \frac{cz}{H_0} = d \\ H_0 &= 70 \text{ km/s/Mpc}\end{aligned}$$

With the goal of deriving ...

$$m - M = 5 \log(z) + 43.2$$

We can do ...

$$\begin{aligned}m - M &= 5 \log_{10}\left(\frac{d}{10 \text{ pc}}\right) \\ m - M &= 5 \log_{10}\left(\frac{\frac{cz}{H_0}}{10 \text{ pc}}\right) \\ m - M &= 5 \log_{10}\left(\frac{\frac{cz}{H_0}}{10^{-5} \text{ Mpc}}\right), \quad m - M = 5 \log_{10}\left(\frac{cz}{H_0} \frac{10^5}{\text{Mpc}}\right) \\ m - M &= 5 \log_{10}\left(\frac{c}{H_0} \frac{10^5}{\text{Mpc}}\right) + 5 \log_{10}(z) \\ 5 \log_{10}\left(\frac{c}{H_0} \frac{10^5}{\text{Mpc}}\right) &\approx 43.15\end{aligned}$$

Hence ...

$$m - M = 5 \log_{10}(z) + 43.2$$

c)

Given ...

$$\begin{aligned}m - M &= 5 \log_{10}(z) + 43.2 \\ Z &= .0023, \quad mV = 12.7\end{aligned}$$

We find ...

$$M_v = -22.379825$$

d)

It is more likely to be an elliptical galaxy, as they have a absolute magnitude of ≈ -22

11.5 Extra Credit

a)

It is a spiral, it has new, ‘infantile’, stars, with the nebulae becoming structured.

b)

Given ...

$$V = H_0 * D$$

$$D = 2,000,000\text{mi}$$

We find ...

$$V = 2.684 \times 10^{-13} \text{ km/s}$$

This means that it is barely moving away from us, whereas the poem says that it is receding 'At several million miles per sec', thus that the author is taking artistic liberty.

c)

It is probably before 1929, because Hubble discovered the proper recession then.